

Introduction to Acids & Bases

Is this your perception of acids & bases?

... some acids & bases can be like this, but most are not.



[Colourful Chemistry of Acids and Bases \(MIT\)](#)

Common Acids



Citric
Acid



Carbo
nic
Acid



Acetic
Acid
(Vinegar)

Substances that you use everyday
can be classified as acids

Note: Not all
acids
are safe to eat!

The opposite of an acid is a base. Example: Soaps

What is an acid?

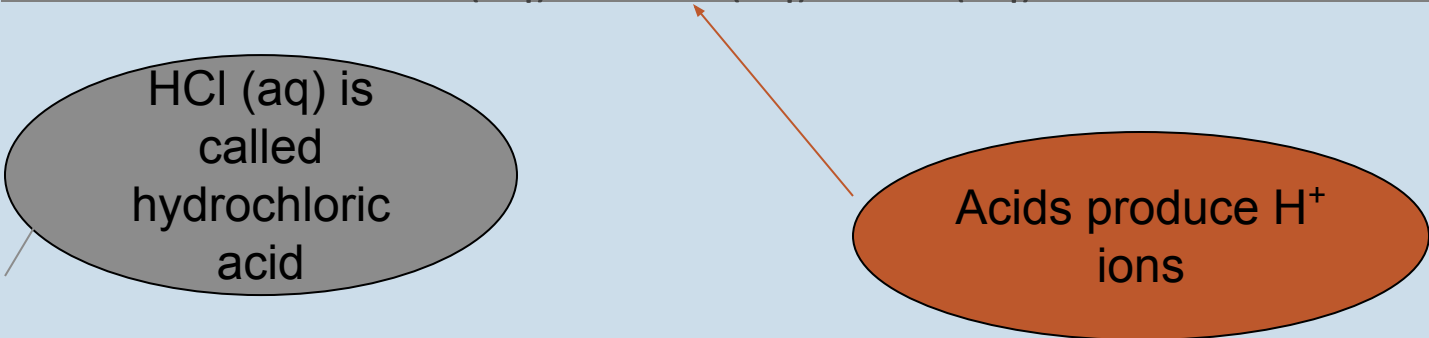
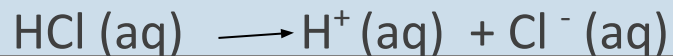
An acid is a compound that produces H^+ ions when dissolved in water.

The symbol (aq) means 'aqueous' or dissolved in water.



Ionization

Hydrogen chloride will **ionize** when dissolved in water:



HCl (aq) is
called
hydrochloric
acid

Acids produce H^+
ions

When ions are in a solution, electricity can be conducted. (The solution is called an electrolyte)

Common Bases

Common Household Acids & Bases



© McGraw-Hill

Acids



Bases

What is one
characteristic of
bases?

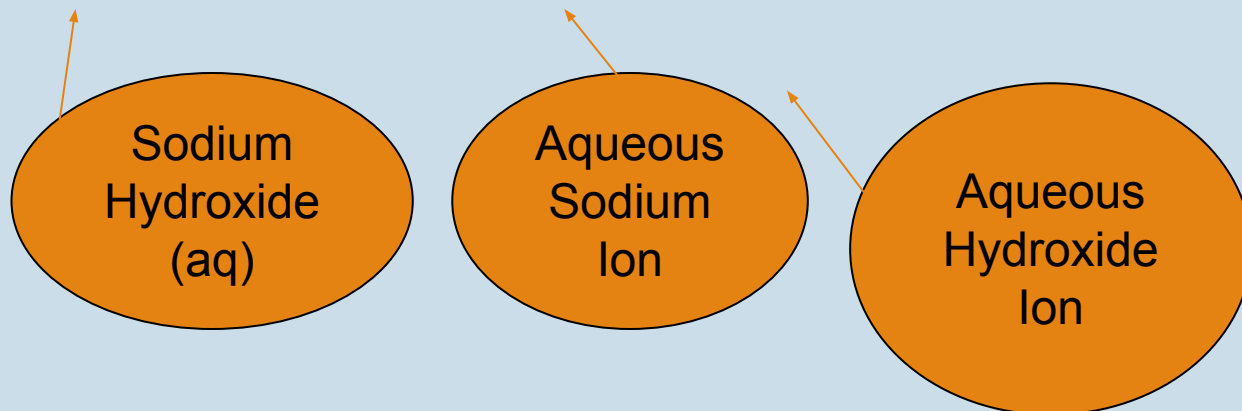
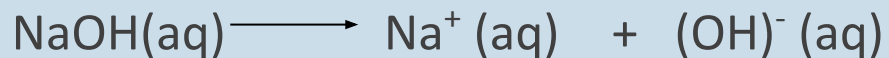
What is a base?

A base is a compound that produces hydroxide (OH^-) ions when dissolved in water.



Dissociation

Aqueous solutions of bases also contain ions and therefore conduct electricity.





Arrhenius definition of acids/bases

Acid - increases concentration of H^+ when in aqueous solution

Bases - increases OH^- concentration when in aqueous solution

* Svante Arrhenius

Strong acid



Arrhenius definition of acids and bases



Khan Academy

Properties Comparison

Property	Acids	Bases
Taste	Sour	bitter
Solubility	Soluble	soluble
Litmus paper result	Acids turn blue litmus paper red	Bases turn Red litmus paper Blue
Electrical	Good conductors Ionize to release H^+ ions in water	Good conductors Ionize to release OH^- ions in water
Reactions	React with some metal to make H_2	React with acids to make salt and water
Found in...	Salad dressings, citrus fruits, aspirin, some pops	Many cleaning products, soaps, baking soda
Formula	Begin with Hydrogen Ex. H_2SO_4	Contain hydroxide or bicarbonate Ex NaOH

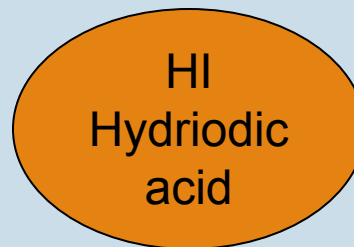
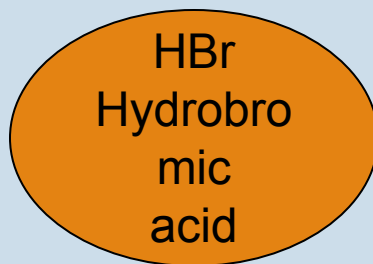
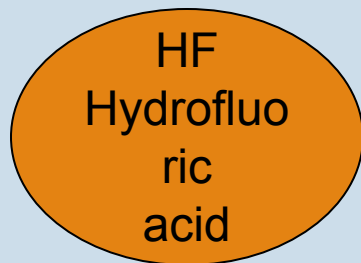
Naming Acids

1. Binary Acids

Composed of 2 elements: hydrogen and a non-metal

Start with 'hydro'

End with 'ic'



Naming Acids

2. Oxoacids

Composed of hydrogen, oxygen, and another element

In general, they are a polyatomic group with a hydrogen cation

NOTE: You use ONLY the name of the polyatomic group and replace endings as below. NO hydro in the name of acids with polyatomic ions.

Replace 'ate' endings
with 'ic acid'

H_2SO_4
Sulfuric
acid

HNO_3
Nitric
acid

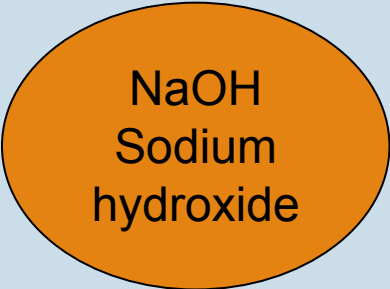
H_3PO_4
Phospho
ric
acid

Replace 'ite' endings
with 'ous acid'

H_2SO_3
Sulfurous
acid

Naming Bases

Either have a common name or just use the standard ionic naming



NaOH
Sodium
hydroxide

What is the pH of a solution if the $[H^{1+}]$ is

1 $\times 10^{-5}$ M?

$$\text{pH} = -\log [H^{+}]$$

$$\text{pH} = -\log [1 \times 10^{-5}]$$

$$\text{pH} = 5$$

If the first number is 1, then the pH is equal to the positive value of the superscript.

To determine $[H^+]$ when given pH, its just reversing the log:

$$[H^+] = 10^{-pH}$$

NOTE that you need to put a negative in front of the pH in the exponent.



don't let anyone judge you...

Strength & Weakness of acids & bases Ted-Ed



ACIDS

AND

BASES

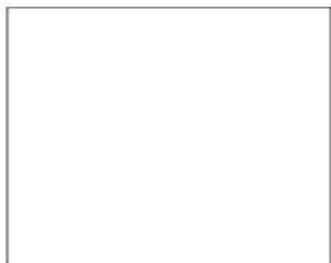
WHAT CAUSES HEARTBURN?



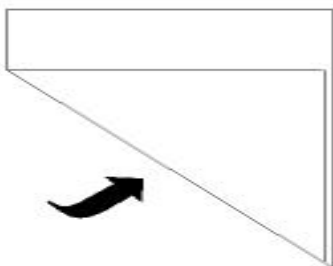
Acid/Base Name Game

Instructions for "Cootie Catcher" and "Fortune Teller"
www.funorama.com

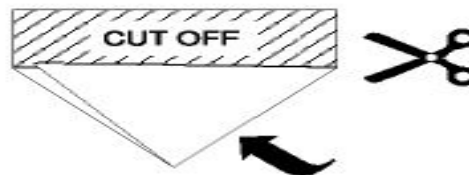
Acid/Base
Name Lists:
p. 222, 223, &
226



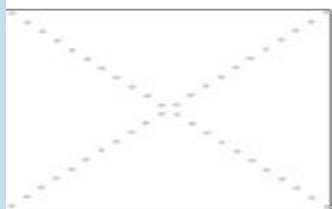
Take an 8 1/2" x 11
sheet of paper"



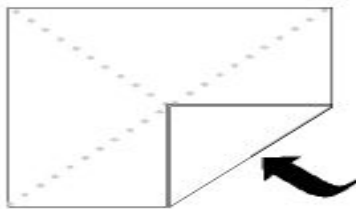
Fold corner up till it meets
the other side



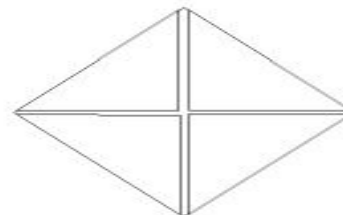
Fold other corner up till it
meets the other side--then
cut off the rectangle at the top.



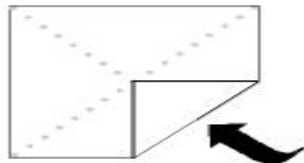
Unfold it and place it
printed side down — you
should now have an
8 1/2" x 8 1/2" piece.



Fold up all four corners
so that the points meet
in the middle.



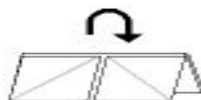
It should now look like this.
Flip it over.



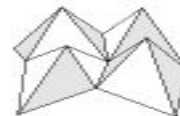
Fold up all four corners
so that the points meet
in the middle.



It should now
look like this.



Now fold the
top back.



Work your fingers into
the four corners from the
fold side — work the
creases to form the
four points.

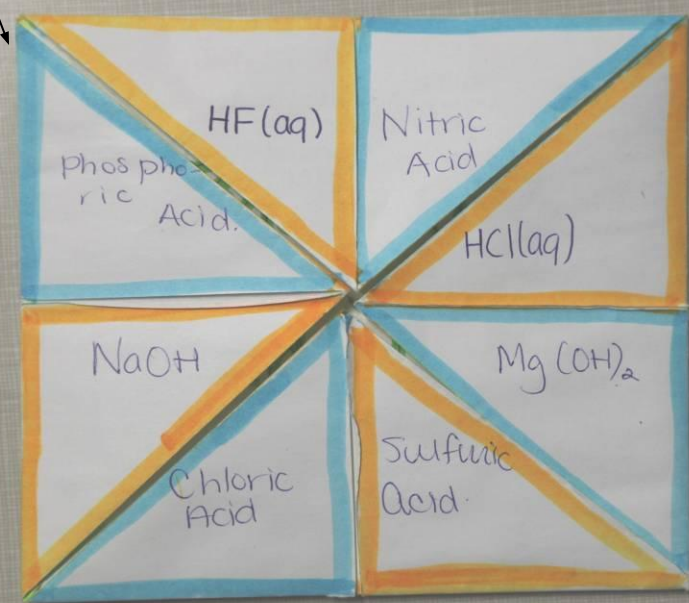
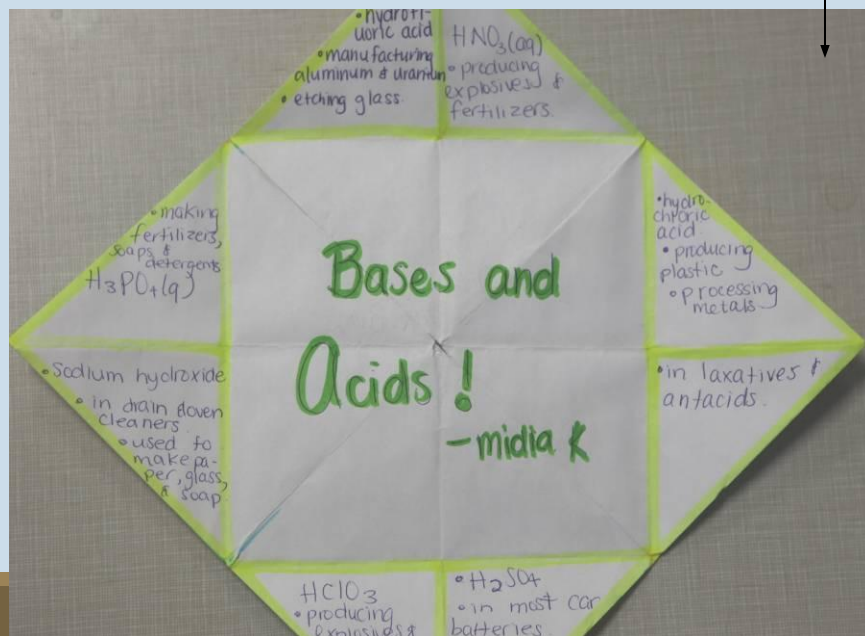
Label names & formulas as follows

1. Place any 4 acid/base words on the outside four 'finger flaps'.

2. i) Choose any 4 acid/base formulas &
ii) Choose any 4 acid/base names ...
and place them on the eight 'inner flaps'.

3. Underneath each 'inner flap' place:
i) the name/formula **answer**;
ii) at least two **different uses**.

See
pages
222
-226



4. Colour code and/or decorate your acid/base 'cootie catcher'.